

CAMERON MONTBRIAND

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SUMMARY

Robotic welding solutions engineer and technical pre-sales lead. I run discovery on customer fabrication floors, build the robot simulation and reach studies that prove an application on the customer's actual parts, and lead the demos that turn a welding problem into a sold system. Daily work in sensor-driven adaptive welding: laser seam tracking and seam finding, touch and arc sensing, and 3D scan-to-program on parts with real fit-up variation. Field territory across 11 states with regular customer travel. I also build the Python and AI tooling that automates my own technical sales workflow.

CORE SKILLS

Robotic Welding Applications: Robotic GMAW and FCAW applications | weld symbology and print interpretation | joint access and reach studies | cycle-time modeling | fixture and positioner concepts | weld test coupon validation | manual welding lab and demo coupon experience

Adaptive Welding and Sensing: Laser seam tracking | laser seam finding | touch sensing | arc sensing | 3D scanning for program generation | offline programming and robot simulation

Technical Pre-Sales: Technical discovery | as-is process and workflow analysis | customer demos and product evaluations | technical presentations | value-based selling and objection handling | cost modeling and quoting | pipeline management (HubSpot CRM)

CAD, Simulation, and Analysis: SolidWorks | AutoCAD | Creo | GD&T and tolerance analysis | FEA and multi-physics (ANSYS, Abaqus) | MATLAB modeling

AI and Automation: Python | Claude and LLM agent workflows | retrieval-augmented generation (Pinecone) | computer-vision document extraction | N8N | REST API integration

PROFESSIONAL EXPERIENCE

CLOOS North America | Schaumburg, IL | Application Engineer, Pre-Sales | March 2024 to Present

Pre-sales engineer on the technical side of every robotic welding system sold in North America. Supported \$19.5M+ in sales.

- Run technical discovery and as-is analysis on customer fabrication floors, mapping upstream and downstream workflow, throughput, and constraints, then translating findings into a tailored automation solution.
- Build robot simulation models and run reach, accessibility, and cycle-time studies on customer weldments, proving feasibility on their real parts before they commit capital.
- Specify sensor-driven adaptive welding packages including laser seam tracking, laser seam finding, touch sensing, arc sensing, and 3D scan-to-program, solving the fit-up variation that defeats fixed-path robots in heavy fabrication.
- Lead customer-facing technical presentations and live demonstrations, running weld test coupons to validate joint access, process, and penetration before purchase.
- Designed \$400K+ in custom automation configurations, tailoring positioners, tooling, and cell layout to nonstandard parts.
- Interpret weld symbols and fabrication prints to scope weld length, process selection, and consumables into cost models, quotes, and compliance documentation.
- Own opportunities from discovery to close in HubSpot, grew a single industrial account to \$12M, and generated 20% of qualified leads through trade shows, onsite visits, and technical discussions.
- Feed field application findings back to engineering to shape product configuration and future capability.

G&W Electric | Bolingbrook, IL | Project Engineer | June 2022 to March 2024

- Designed and tested mechanical actuating systems for medium-voltage switchgear to IEEE standards, running structured test matrices to validate mechanism performance.
- Directed field repair and implementation across 200+ installed systems, closing the loop between field failure and design change and improving product reliability.
- Ran GD&T and tolerance analyses that improved manufacturability and reduced defect rates on welded and machined assemblies.

SELECTED AI AND AUTOMATION PROJECTS

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- BOM Creator (Python, vision model): reads steel fabrication drawings and generates a structured bill of materials in Excel, replacing manual takeoff from prints.
- AI Service Chatbot (Claude, Pinecone, RAG): indexed 111 welding service manuals so technicians get step-by-step fault resolution in plain English instead of searching PDFs.
- AI Sales Prospecting System (Python, Claude Code, HubSpot, Hunter.io): five-agent pipeline that qualifies companies and maps decision makers across an 11-state territory, removing 10 hrs/week of manual research.
- CLOOS Trainer (Claude Code): 84 interactive training modules built from 111 service manuals, replacing \$10K+ in vendor-quoted production cost.

EDUCATION

University of Iowa | Iowa City, IA | Bachelor of Science, Mechanical Engineering | May 2022 | GPA 3.54

Dean's List, 3 semesters. Hands-on welding lab coursework; FEA (ANSYS, Abaqus), MATLAB modeling.